

Data Cleaning and Visualization using Titanic Dataset

Internship: Data Science Internship

Task: Data Cleaning & Visualization

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Objective

To clean the Titanic dataset by handling missing values, checking duplicate records, and visualizing the data using Python.

Introduction

Data cleaning improves data quality by handling missing values and inconsistencies. Data visualization helps identify patterns and trends.

Dataset Description

Dataset: Titanic

Rows: 891

Columns: 12 (before dropping Cabin)

Tools: Google Colab, Python

Libraries: Pandas, Matplotlib, Seaborn

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	NaN	S
1	2	1	1	Cummings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833	C85	C
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	NaN	S
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000	C123	S
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0500	NaN	S

Data Cleaning

1. Loaded dataset.
2. Checked dataset information using `df.info()`.

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
 #   Column          Non-Null Count  Dtype  
---  -
 0   PassengerId     891 non-null    int64  
 1   Survived        891 non-null    int64  
 2   Pclass          891 non-null    int64  
 3   Name            891 non-null    object  
 4   Sex             891 non-null    object  
 5   Age             714 non-null    float64 
 6   SibSp           891 non-null    int64  
 7   Parch           891 non-null    int64  
 8   Ticket          891 non-null    object  
 9   Fare            891 non-null    float64 
10   Cabin           204 non-null    object  
11   Embarked        889 non-null    object  
dtypes: float64(2), int64(5), object(5)
memory usage: 83.7+ KB
```

3. Checked missing values using `df.isnull().sum()`.

PassengerId	0
Survived	0
Pclass	0
Name	0
Sex	0
Age	177
SibSp	0
Parch	0
Ticket	0
Fare	0
Cabin	687
Embarked	2

dtype: int64

4. Filled Age with median.

5. Filled Embarked with mode.

6. Dropped Cabin column.

7. Verified no missing values remain.

PassengerId	0
Survived	0
Pclass	0
Name	0
Sex	0
Age	0
SibSp	0
Parch	0
Ticket	0
Fare	0
Embarked	0

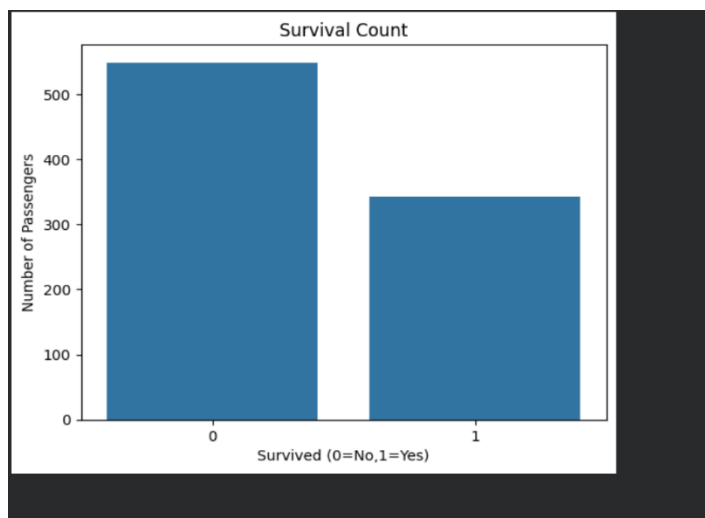
dtype: int64

8. Checked duplicates using `df.duplicated().sum() = 0`.

```
df.duplicated().sum()  
  
np.int64(0)
```

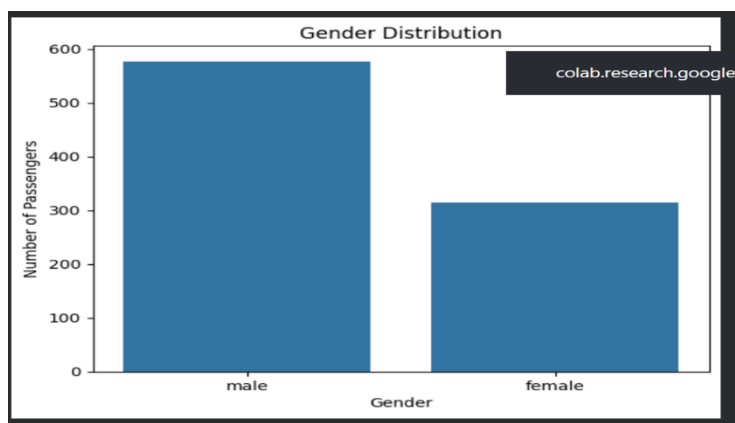
Data Visualization

Survival Count



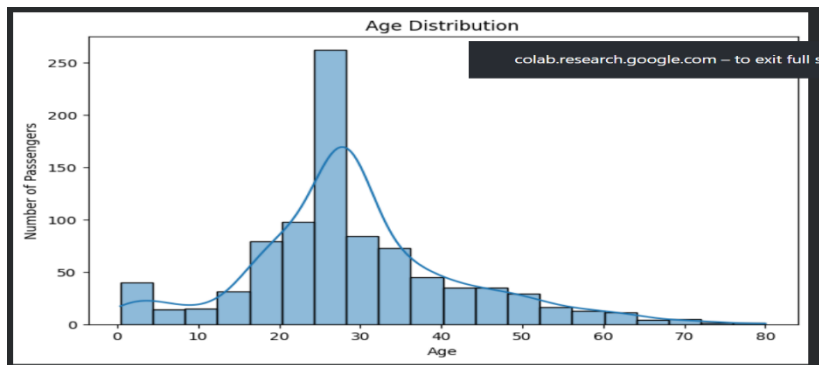
The Survival Count graph shows the number of passengers who survived and those who did not survive in the Titanic disaster. It can be observed that the number of passengers who did not survive was higher than the number of passengers who survived. This provides an overview of the survival distribution in the dataset

Gender Distribution



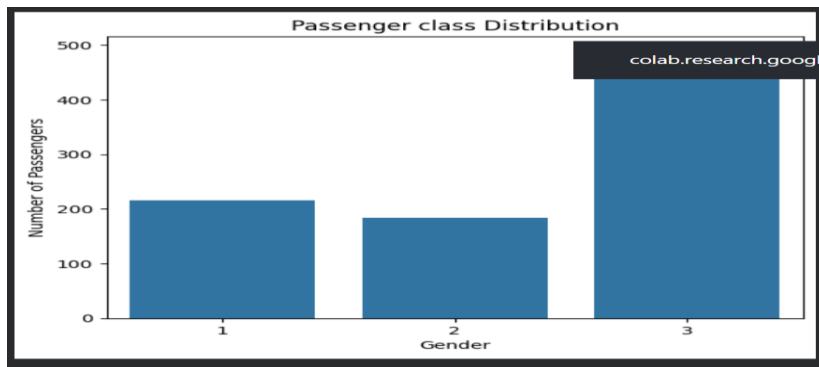
The graph shows the distribution of male and female passengers in the Titanic dataset. It indicates that the number of male passengers was higher than the number of female passengers

Age Distribution



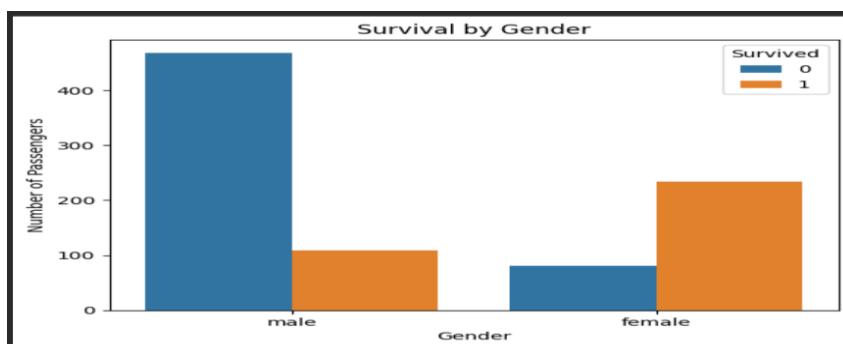
The histogram shows the distribution of passengers based on their age. Most passengers were young adults, while fewer passengers belonged to older age groups.

Passenger Class Distribution



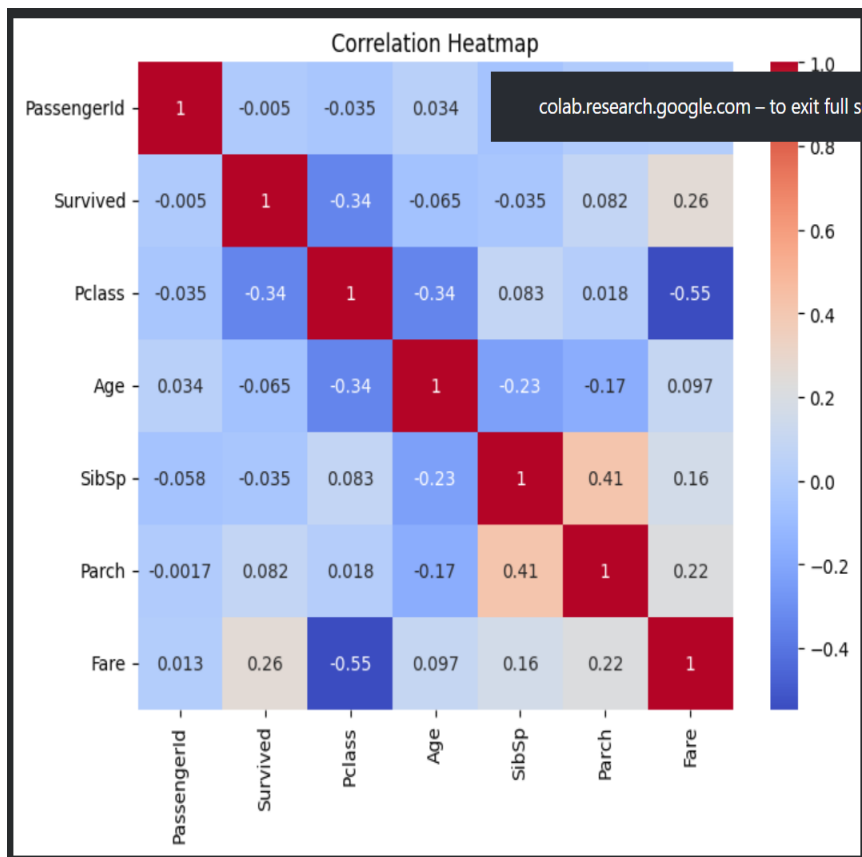
The graph shows the number of passengers traveling in each passenger class. It indicates that the highest number of passengers traveled in the **third class**, followed by the first and second classes.

Survival by Gender



The graph compares the survival of male and female passengers. It shows that female passengers had a higher survival rate than male passengers.

Correlation Heatmap



The correlation heatmap shows the relationship between the numerical features in the dataset. Stronger correlations are represented by darker colors, while weaker correlations are shown by lighter colors.

Results

The dataset was cleaned successfully. Missing values were handled, duplicate records were checked, and multiple visualizations were created.

Conclusion

The Titanic dataset was successfully cleaned and visualized. The project demonstrates essential data preprocessing and visualization techniques used in data science.